MILITARY SPECIFICATION
PROJECTILE, 105MM: SMOKE, BE, M84E1
METAL PARTS ASSEMBLY

1. SCOPE

1.1 This specification covers Projectile, 105MM: SMOKE, BE, M84E1, Metal Parts Assembly.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-A-2550 - Ammunition and Special Weapons, General Specification for

STANDARDS

Federal

Federal Test Method Standard No. 141 - Paint, Varnish, Lacquer and Related Materials, Method of Inspection, Sampling and Testing.

Federal Test Method Standard No. 151 - Metals; Test Methods

Military

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

MIL-STD-109 - Quality Assurance Terms and Definitions

MIL-STD-129 - Marking for Shipment and Storage

FSC 1315

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2.1 (Cont'd)

MIL-STD-1167 - Ammunition Data Card
MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunition Components
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

Munitions Command
D10542988 - Projectile, 105MM, SMOKE, BE, M84E1, Metal Parts Assembly
IEL-10542988 - Inspection Equipment List for Projectile, 105MM, SMOKE, BE, M84E1, Metal Parts Assembly
F8837835 - Pallet for Fixed and Semi-Fixed Shell
D8593429 - Band Seating Testing Machine Indentor and Test Data

PUBLICATIONS

MUCOM
Frankford Arsenal
Operating instructions for use of Detroit Testing Machine for Band Seat Inspection

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other Publications.-The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

ASA-B46.1 - American Standards Surface Texture, Surface Roughness, Waviness and Lay

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

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3. REQUIREMENTS

3.1 General.-Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings, referenced specifications and the following:

3.2 Heat treatment or stress relief.-Heat treatment or stress relief applied to the projectile body shall be uniform throughout and shall be so conducted as to produce in each projectile body the mechanical properties specified by the applicable drawing. Metal forming involving plastic flow without subsequent heat treatment of stress relief is not permitted.

3.2.1 Hot Forged- Heat treatment method of manufacture.-Projectile bodies manufactured from forgings shall be subjected to heat treatment to attain the mechanical properties specified by the applicable drawing. Heat treatment shall consist of heating above the critical temperature, not to exceed 1600 degrees Fahrenheit (°F), followed by an oil quench. After quenching, the body forging shall be tempered without delay at a temperature above 750 °F which will produce the required mechanical properties. After removal from the tempering furnace, the body forging shall be slowly cooled in air (forced air permitted) to a temperature of not more than 700 °F, after which cooling may be accelerated by water spray.

3.2.2 Cold shaping and hot cup - Cold draw method of manufacture.-Cold working shall be so accomplished as to produce in the projectile body the mechanical properties specified by the applicable drawing. The projectile body so formed shall be stress relieved at a temperature of 700 °F, minimum, following the final forming operation which results in plastic flow. The minimum soak time at or above minimum stress relief temperature shall be established which will produce mechanical properties specified by the applicable drawing. The temperature and time duration of in process anneals prior to the final forming operation shall be predicated on attainment of required mechanical properties when formed by an approved sequence of operations.

3.3 Mechanical properties.

3.3.1 Projectile body.-Mechanical properties specified on the applicable drawing shall apply throughout the projectile body.

3.3.2 Rotating band.-After application and subsequent removal from the projectile body, the rotating band shall have the following mechanical properties:

Tensile Strength - 30,000 pounds per square inch (PSI) minimum
Elongation - 15 percent minimum in 1 inch

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3.4 Rotating band seating.

3.4.1 Band seat cleanliness.-Prior to banding, the band seat shall be clean and free of oil, dirt, grease, rust or other foreign matter. Shot or sand blasting is not permitted for cleaning the band seat after knurling.

3.4.2 Application of band blank.-The rotating band blank shall be pressed into the band seat, after final heat treatment of the projectile body. The band blank shall be applied in such manner as not to distort the projectile body. The band blank must contact the sidewall of the band seat for a minimum of eight-tenths of the band seat depth measured from the outside surface of the projectile body. There shall be no gap between the band and the band seat sidewalls at the outside surface of the projectile body after the rotating band has been machined to final dimensions.

3.4.3 Rotating band clearance.-The diametrical clearance between the rotating band and the band seat shall not exceed .006 inch.

3.5 Metal defects.-All components shall be free from cracks, seams, splits, cold shuts, inclusions, lamination, porosity or similar defects.

3.6 Protective coating.

3.6.1 Body assembly and base plug.-Requirements relative to paint thickness are not specified except as a guide to performance of tests as indicated by the applicable drawings and specifications. Actual component parts must withstand the specified salt spray exposure with coating breakdown not greater than that described by the applicable specification.

3.6.2 Baffle plate.-All aspects of surface pretreatment or surface coating are as shown by the applicable specification, and are subject to demonstration of control of process thru salt spray test of panels as indicated by the applicable specification.

3.7 Functional.

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3.7.1 Metal parts security.-When tested in accordance with 4.4.5. The projectile assembly shall reveal no evidence of metal parts failure. Metal parts failure is defined as:

1. Projectile body or base plug deformation attributable to noncompliance with drawing or specification requirements.
2. Missing or loose rotating band attributable to noncompliance with drawing or specification requirements.
3. Shear of rotating band as evidenced by complete lack of engraving.

3.8 Workmanship.-The requirements for workmanship are as shown on the applicable drawings, in the applicable specifications and the following:

3.8.1 Surface finish.-The requirements for surface finish are as detailed on the applicable component drawing.

3.8.2 Protective coating.-The protective coating shall be in accordance with the applicable drawing and the specifications listed theron.

3.8.3 Threads.-Threads shall be full and undamaged for the entire minimum length or depth as required by the applicable drawing and shall be in accordance with the provisions of MIL-A-2550.

3.8.4 Burr.-No part shall have a burr which might interfere with the assembly or function of the round or which might be injurious to personnel handling the item.

3.8.5 Foreign matter.-No part or assembly shall contain dirt, grease, chips, oil, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.-Reference shall be made to MIL-STD-109 to define quality assurance terms used.

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4.1.2 Inspection.-Inspection shall be in accordance with MIL-A-2550, except as specified herein.

4.2 First article sample.

4.2.1 Initial production sample.-At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of forty metal parts as follows:

30 Metal Parts Assemblies, complete, including paint
5 Metal Parts Assemblies, complete, except paint
5 Sets of Component Parts, unpainted, unassembled, including rotating band blanks. The bodies shall be completely machined, including the rotating band seat.

Twenty-five of the above complete and painted metal parts assemblies shall be forwarded to a designated proving ground for ballistic testing. The sample shall be manufactured using the same materials, equipment, processes and procedures, including packaging and packing, as will be used in regular production and shall be obtained from the same source of supply.

4.2.1.1 Examination and test.-After inspection and provisional acceptance at source, the sample shall be inspected for all requirements of the drawings and specifications at a Government laboratory or such other facility specified in the contract.

4.2.1.2 Initial production sample failure.-Failure of the sample to comply with the requirements of the drawings and specifications shall result in sample disapproval.

4.3 Inspection provisions.

4.3.1 Lot.

4.3.1.1 Submission of product.-The product shall be submitted in accordance with MIL-STD-105.

4.3.1.2 Heat treatment lots.-Heat treatment lots shall contain body forgings from not more than one heat of steel that have been heat treated in the same heat treating equipment, at the same temperature in one unchanged process. The heat treatment lot thus constituted shall be the basis for the selection of samples for the tension test hereinafter prescribed.

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4.3.1.3 Stress relieved lots.-Stress relieved lots shall contain cold shaped bodies from not more than one heat of steel that have been cold shaped by the same process, utilizing the same tooling and equipment and are stress relieved at the same temperature for the same length of time. The stress relieved lot thus constituted shall be the basis for the selection of samples for the tension test hereinafter prescribed.

4.3.1.4 Miscellaneous lots.-Small lots (such as bodies manufactured from abnormally small heats of steel or bodies that have become separated from previously delivered lots, etc.) may at the discretion of the supplier by grouped into miscellaneous heat treatment or stress relieved lots as the case may be. Such lots shall be identified by a miscellaneous lot code number and processed in accordance with either 4.3.1.2 or 4.3.1.3, as applicable. The miscellaneous lot thus constituted shall be the basis for the selection of samples for the tension test hereinafter prescribed.

4.3.1.5 Projectile lots.-When it has been determined that bodies of a heat treated, stress relieved or miscellaneous lot have complied with the mechanical properties requirement of the applicable drawing, they may be re-grouped for assembly into projectile lots, to which will be assigned a projectile lot number in accordance with MIL-STD-1168. The projectile lot shall consist of projectile assemblies produced by one supplier under one contract in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision, under conditions of continuous production free from interruptions other than those due to the end of a shift, day, or workweek.

4.3.2 Examination.-One hundred percent examination shall be performed for all critical defects. Examination for Major and Minor shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plans contained in MIL-STD-105. An AQL of 0.40 percent for each Major defect and an AQL of 0.65 percent for each Minor defect shall be used. Continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the procuring activity. All nonconforming material shall be rejected. Four power (4X) magnification may be used in performing visual examination for such defects as cracks, burr and foreign matter. The roughness comparison specimens prescribed by ASA-B46.1 shall be used as a basis for surface roughness determination.

4.3.2.1 Body, Drawing D10542990 covering a detail of Drawing D10542988

Defect	Method of Inspection
CRITICAL	
1. Metal defective	Visual

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4.3.2.1 (Cont'd)

Defect	Method of Inspection
MAJOR	
101. Diameter of rotating band seat	Gage
102. Width of rotating band seat	Gage
103. Ribs missing from band seat	Visual
104. Scoring missing from band seat	Visual
105. Evidence of sand or shot blast on band seat after scoring	Visual
MINOR	
201. Location of band seat	Gage
202. Foreign matter on band seat	Visual

4.3.2.2 Body assembly (prior to painting) Drawing D10542989 covering a detail of Drawing D10542988

CRITICAL	
1. Open band (gap between band and band seat wall)	Visual
MAJOR	
101. Weight	Approved Scale
102. Concentricity of pitch diameter of nose thread with bourrelet, including perpendicularity of front face	Gage
103. Concentricity of ogive with bourrelet	Gage
104. Concentricity of body with bourrelet	Gage
105. Concentricity of rotating band flat with bourrelet	Gage
106. Concentricity of boat tail with bourrelet	Gage
107. Concentricity of pitch diameter of rear thread with bourrelet	Gage
108. Base plug cavity not functionally concentric with minor diameter of rear thread	Gage
109. Diameter of bourrelets, min.	Gage
110. Diameter of large cavity	Gage
111. Diameter of small cavity rear of forward thread	Gage
112. Diameter rear of small cavity, min.	Gage
113. Diameter of body, outside	Gage
114. Diameter of base plug cavity	Gage

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4.3.2.2 (Cont'd)

	Defect	Method of Inspection
115.	Length to ogive basic diameter	Gage
116.	Length to basic diameter on boattail	Gage
117.	Profile of basic angle on boattail	Gage
118.	Thickness of wall at small cavity, min.	Gage
119.	Variation of wall thickness	Gage
MINOR		
201.	Diameter of nose	Gage
202.	Diameter of rotating band relief groove	Gage
203.	Depth to rear of small cavity	Gage
204.	Depth of base plug cavity	Gage
205.	Length to rear of forward bourrelet	Gage
206.	Length to counterbore at rear of small cavity	Gage
207.	Length of counterbore at rear of small cavity, max. (if applicable)	Gage
208.	Location of rotating band	Gage
209.	Length, total	Gage
210.	Profile of rotating band	Gage
211.	Profile of rotating band relief groove	Gage
212.	Width of rotating band	Gage
213.	Depth of undercut at front of rotating band, max. (if applicable)	Visual (see 6.1)
214.	Depth of undercut at rear of rotating band, max. (if applicable)	Visual (see 6.1)
215.	Width of undercut at front of rotating band, max. (if applicable)	Visual (see 6.1)
216.	Width of undercut at rear of rotating band, max. (in applicable)	Visual (see 6.1)
217.	Surface finish, improper (except as otherwise classified)	Visual (see 6.1)
218.	Radii or chamfers, missing or incorrect	Visual (see 6.1)
219.	Burr	Visual
220.	Foreign matter	Visual

4.3.2.3 Plug base Drawing C10542992 covering a detail of Drawing D10542988

CRITICAL

None defined

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4.3.2.3 (Cont'd)

Defect	Method of Inspection
MAJOR	
101. Major diameter of thread not functionally concentric with outside diameter	Gage
102. Pitch diameter of thread, min.	Gage
103. Pitch diameter of thread, max.	Gage
104. Major diameter of thread, min.	Gage
105. Diameter large	Gage
106. Length of thread	Gage
MINOR	
201. Distance between wrench holes	Gage
202. Length, total	Gage
203. Depth of drill point, max. (if applicable)	Gage
204. Presence of wrench holes	Visual (see 6.1)
205. Surface finish, improper	Visual (see 6.1)
206. Radii or chamfers, missing or incorrect	Visual (see 6.1)
207. Corrosive preventive compound missing in indicated areas	Visual
208. Protective coating inadequate or damaged	Visual
209. Burr	Visual
210. Foreign matter	Visual

4.3.2.4 Gasket, Drawing C10542991 covering a detail of Drawing D10542988

CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Diameter large	Gage
202. Diameter small	Gage
203. Thickness	Gage
204. Burr	Visual
205. Foreign matter	Visual

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4.3.2.5 Plate, Baffle, Drawing C10542993 covering a detail of Drawing D10542988

Defect		Method of Inspection
CRITICAL		
	None defined	
MAJOR		
101.	Diameter large	Gage
102.	Diameter of hole	Gage
103.	Thickness	Gage
MINOR		
201.	Surface finish, improper	Visual (see 6.1)
202.	Radii or chamfers, missing or incorrect	Visual (see 6.1)
203.	Protective coating inadequate or damaged	Visual
204.	Burr	Visual
205.	Foreign matter	Visual

4.3.2.6 Metal Parts Assembly, Drawing D10542988

CRITICAL		
	None defined	
MAJOR		
101.	Pitch diameter of forward thread, max.	Gage
102.	Pitch diameter of forward thread, min.	Gage
103.	Pitch diameter of rear thread, max.	Gage
104.	Pitch diameter of rear thread, min.	Gage
105.	Minor diameter of forward thread, max.	Gage
106.	Minor diameter of rear thread, max.	Gage
107.	Diameter of bourrelets, max.	Gage
108.	Length of effective forward thread, min.	Gage
109.	Length of rear thread, min.	Gage
110.	Paint on threads	Visual
111.	Paint on base plug cavity surface	Visual
112.	Rotating band damaged	Visual
113.	Corrosive preventive compound missing in indicated areas	Visual

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4.3.2.6 (Cont'd)

Defect	Method of Inspection
MINOR	
201. Staking notches missing or incorrect	Visual (see 6.1)
202. Protective coating inadequate or damaged	Visual
203. Marking missing, incorrect or illegible	Visual
204. Foreign matter	Visual
205. Protective cap(s) missing	Visual

4.3.2.7 Pallet, Drawing F8837835

CRITICAL

None defined

MAJOR

None defined

MINOR

201. Pallet damaged (unserviceable)	Visual
202. Sleeve missing	Visual
203. Improperly packed	Visual
204. Strapping missing, loose or broken	Manual/Visual
205. Marking missing, incorrect or illegible	Visual

4.3.3 Inspection testing.-The following tests shall be performed in accordance with the provisions of 4.4:

4.3.3.1 Mechanical properties.

4.3.3.1.1 Hardness.

4.3.3.1.1.1 Heat treatment lots.-In order to determine uniformity of heat treatment, each body forging in the heat treatment lot (as defined by 4.3.1.2) shall be subjected to a hardness test with a brinnell hardness tester, using a 5-second minimum dwell or by any other method of hardness testing approved by the procuring activity.

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4.3.3.1.1.2 Stress relieved lots.-Twenty-five cold shaped bodies from the first batch of each stress relieved lot (as defined by 4.3.1.3) processed through the stress-relief furnace and five bodies from each 1-hour production thereafter for the balance of the stress relieved lot shall be subjected to a hardness test to assess uniformity of the process.

4.3.3.1.1.3 Miscellaneous lots.-Miscellaneous heat treatment lots shall be subjected to a hardness test in accordance with 4.3.3.1.1.1. Miscellaneous stress relieved lots shall be subjected to a hardness test in accordance with 4.3.3.1.1.2.

4.3.3.1.2 Tension test.

4.3.3.1.2.1 Heat treatment lots.-Two projectile body forgings, one of the hardest and one of the softest as determined by 4.3.3.1.1.1 shall be subjected to this test. Failure of any tension test specimen to comply with the yield strength or elongation requirements of the applicable drawing shall be cause for rejection of the lot. At the option of the supplier, a production control sample may be utilized to establish a hardness range. When a production control sample is used, the hardness range thus established shall be maintained throughout the heat treatment lot. The size of the sample shall be determined by the contractor. Projectile bodies which fall outside of the established range may be re-heat treated to fall within the range, or new mechanical properties may be determined from newly selected projectile bodies which will represent the true extremes of hardness in the heat treatment lot.

4.3.3.1.2.2 Stress relieved lots.-Two projectile bodies from each stress relieved lot (as defined by 4.3.1.3) shall be selected at random and subjected to this test. Failure of any tension test specimen to comply with the yield strength or elongation requirements of the applicable drawing shall be cause for rejection of the lot.

4.3.3.1.2.3 Miscellaneous lots.-Samples for tension test of miscellaneous heat treatment or stress relieved lots shall be selected in accordance with 4.3.3.1.2.1 or 4.3.3.1.2.2, as applicable. Failure of any tension test specimen to comply with the yield strength or elongation requirements of the applicable drawing shall be cause for rejection of the lot. Double the number of samples shall be selected for miscellaneous lots.

4.3.3.1.2.4 Rotating band.-One body assembly shall be selected from each 8-hour production or fraction thereof from each banding machine in operation and subjected to this test to determine compliance with 3.3.2. Failure to comply shall be cause for rejection of the assemblies represented by the sample.

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4.3.3.2 Band seating.

4.3.3.2.1 Tightness.

4.3.3.2.1.1 Non-destructive method.-Five body assemblies from each 1-hour production per banding machine in operation shall be subjected to this test. Failure of one or more bands to comply with 3.4.3 shall be cause for rejection of the assemblies represented by the failing sample.

4.3.3.2.1.2 Destructive method.-One body assembly from each 8-hour production or fraction thereof per banding machine in operation shall be subjected to this test to determine compliance with 3.4.3. Failure to comply shall be cause for rejection of the assemblies represented by the failing sample.

4.3.3.3 Cleaning and application of protective coating.-The sample size and acceptance criteria for inspection of the cleaning of the item and the application of protective coating shall be as specified in the applicable specification. Failure to comply with the requirements shall be cause for taking the corrective action prescribed therein.

4.3.4 Straightness inside diameter (a major defect).-The inside diameter of each body assembly (after banding) shall be gaged as prescribed by the applicable drawing. Failure of the functional gage to enter for the full indicated depth shall be cause for rejection of the body assembly.

4.3.5 Functional.-The following Proving Ground tests will be performed by the Government:

4.3.5.1 Security.-Twenty-five metal parts assemblies from the initial production sample will be ballistically tested to determine compliance with 3.7.1. Failure of any projectile to comply will be cause for rejection of the sample.

4.3.6 Inspection equipment.-Index of Inspection Equipment Lists IEL-10542988 identifies the inspection equipment required to perform the examinations and tests prescribed herein.

4.4 Test methods and procedures.

4.4.1 Mechanical properties.

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4.4.1.1 Hardness.-After final heat treatment but prior to machining, the brinnell hardness test shall be performed on the outside surface at the band seat location. The test shall be performed using the equipment and procedures for performing this test as prescribed by Federal Test Method Standard No. 151, except that a 5-second minimum dwell shall be used. A direct reading hardness tester may be substituted for the Brinnell scope (micrometer miscroscope) type, provided that the same size indenter ball and the same pressure is used and that 2000 body forgings will have been tested by both units to establish correlation. Continued correlation shall be assured by checking the hardness value of one body forging each hour by measuring the diameter of the indent with a Brinnellscope, for comparison with the dial reading.

4.4.1.2 Tension test.

4.4.1.2.1 Body.-From the samples provided by 4.3.3.1.2.1, 4.3.3.1.2.2 or 4.3.3.1.2.3 for the convenience of the Government two tension test specimens conforming to the largest obtainable standard "R" type specimen, as prescribed by Federal Test Methods Standard No. 151, shall be taken from opposite sides of the body, at mid-wall in such a manner that the center of the gage length of the machined specimen will fall in line coincident with the center of the rotating band seat, or from any other location in the body deemed necessary by the Government. The criteria for performing this test shall be as prescribed by Federal Test Methods Standard No. 151, yield strength shall be determined by the offset method the limiting set shall be .01 percent (0.001 inch per inch of gage length).

4.4.1.2.2 Rotating band (gilding metal).-The samples provided by 4.3.3.1.2.4 shall have the rotating band removed and carefully flattened without hammering so as to hold to a minimum any additional cold working. The specimen blank shall be machined so that all serration marks from the band seat are removed. A test specimen having a one inch gage length and a one-half inch gage width, and grip ends of sufficient length to accomodate the tensile machine shall be removed from the blank, and tested in accordance with the procedures prescribed by Federal Test Method Standard No. 151.

4.4.2 Band seating

4.4.2.1 Non-destructive method.-The samples provided in 4.3.3.2.1.1 shall be tested utilizing the equipment and procedures prescribed in Frankford Arsenal Detroit Testing Machine Manual (see 6.2) and line 19 of Drawing D8593429 Rev. C.

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4.4.2.2 Destructive method.-With the approval of the procuring activity when the Detroit Testing Machine is not available, or is not functioning correctly the destructive method may be used in lieu thereof. Rotating bands of the sample provided in accordance with 4.3.3.2.1.2 need not be processed through the band machineing line, although the band should have a light surface cut (.020 max.) to true up the surface prior to measuring. The diameter of each band shall be measured on 3 locations, 60 degrees apart, on 2 points along the cylindrical portion of their length, near the front and rear edges. This comprises 6 diametrical measurements per band; these points of measurement shall be marked on the bands and on the projectile body adjacent thereto in such a manner as to permit proper body and band orientation after removal. The bands shall be carefully removed so as to keep deformation to a minimum. After removal, the thickness of the band shall be measured at each of the marked points (a total of 12 measurements) and the diameter of the band seat corresponding to these points shall also be measured. Add to each of the diameters of the band seat, the sum of the corresponding 2 thicknesses of the band; the results obtained shall then be subtracted from the previously determined diameters over the band. The 6 results shall be totaled and the sum thus obtained will be divided by 6 to determine the average diametrical clearance.

4.4.3 Inside diameter.-The functional gage shall be entered into the cavity of the body assembly to the depth indicated on the applicable drawing.

4.4.4 Protective coating.-The procedure for testing the cleaning of components and the application of protective coating shall be in accordance with the criteria for testing, as specified in the applicable specification. The salt spray test shall be performed utilizing the procedures and equipment prescribed by Federal Test Method Standard No. 141.

4.4.5 Functional.

4.4.5.1 Security.-The metal parts assemblies supplied in accordance with 4.3.5.1 inert loaded to the service weight will be assembled into complete rounds and fired in a gun for which it is standard, the propellant charge will be adjusted to obtain 112 percent of the service pressure of the round in a new tube. The projectile will be fired into a recovery area. An attempt will be made to recover at least 60 percent of the projectile fired. Each recovered projectile will be examined to determine compliance with 3.7.1.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking.-Level C - Packing and marking shall comply with the applicable drawings and specifications, MIL-STD-1169 and the following:

5.2 Packing.

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5.2.1 Body assembly.-Body assemblies, part No. 10542990, with cap protectors in place, shall be palletized in accordance with Drawing F8837835.

5.2.2 Base plug.-Base plugs, part No. 10542992 shall be packed in accordance with MIL-STD-1169. Pack twenty-four (24) base plugs, arranged in a 2 x 2 pattern, 1 to a cell in 6 layers using interlocking separators.

5.2.3 Gasket.-Gasket, part No. 10542991, shall be packed in accordance with MIL-STD-1169. Pack five hundred (500) gaskets, arranged in a 2 x 2 pattern, 125 to a cell using interlocking separators.

5.2.4 Plate, Baffles.-Baffle plate, part No. 10542993 shall be packed in accordance with MIL-STD-1169. Pack one hundred (100) baffle plates, arranged in a 2 x 2 pattern, 25 to a cell using interlocking separators.

5.3 Marking.

5.3.1 Body assembly.-Palletized body assemblies shall be marked in accordance with drawing F8837835.

5.3.2 Base plug, gasket and baffle plate.-Packaged base plugs, gaskets and baffle plates shall be marked in accordance with the marking requirements of MIL-STD-1169.

5.4 Shipping.-When component parts from more than one lot are shipped as a carload, each lot must be kept separated and division between lots clearly indicated to prevent mixing in transit.

5.5 Data cards.-The contractor shall prepare and furnish data cards in the number to be determined by the contracting officer in accordance with MIL-STD-1167.

6. NOTES

6.1 Visual examination qualification.-When compliance with the applicable requirements is in doubt as a result of visual examination, this characteristic may be measured or gaged to determine acceptability.

6.2 Band seating.-The operating procedures for use of Detroit Testing Machine as prepared by the Inspection Egnineering Office, U.S. Army, Frankford Arsenal, may be used as an aid in checking band seating inspection procedures.

Custodian:

Army - MU

Preparing Activity

Army - MU

Project No. 1315-A648

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

[ ] YES [ ] NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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